

MySQL Locking Blocking & Deadlock Scenario

1. Locking

Create database and table .

```
CREATE DATABASE IF NOT EXISTS lock_test;
```

```
USE lock_test;
```

```
CREATE TABLE employee (  
    id INT PRIMARY KEY,  
    name VARCHAR(100),  
    salary DECIMAL(10,2)  
) ENGINE=InnoDB;
```

```
INSERT INTO employee (id, name, salary)  
VALUES  
(1, 'John', 50000),  
(2, 'David', 60000),  
(3, 'Mike', 70000);
```

Open session 1 to login mysql :

```
USE lock_test;
```

```
START TRANSACTION;
```

```
UPDATE employee  
SET salary = 55000  
WHERE id = 1;
```

Note : Do not run Commit or Rollback

Open session 2
Mysql -u root -p

```
USE lock_test;
```

```
UPDATE employee  
SET salary = 65000  
WHERE id = 1;
```

Note : Now session 2 is hang / wait

Note : after this below query use to find the blocking

```
SELECT  
    r.trx_id AS waiting_trx_id,  
    r.trx_mysql_thread_id AS waiting_thread,  
    r.trx_query AS waiting_query,  
    b.trx_id AS blocking_trx_id,  
    b.trx_mysql_thread_id AS blocking_thread,  
    b.trx_query AS blocking_query  
FROM information_schema.innodb_trx r  
JOIN performance_schema.data_lock_waits w  
    ON r.trx_id = w.REQUESTING_ENGINE_TRANSACTION_ID
```

```
JOIN information_schema.innodb_trx b
  ON b.trx_id = w.BLOCKING_ENGINE_TRANSACTION_ID;
```

Next See exactly what is locked .

```
SELECT
  ENGINE_TRANSACTION_ID,
  THREAD_ID,
  OBJECT_SCHEMA,
  OBJECT_NAME,
  INDEX_NAME,
  LOCK_TYPE,
  LOCK_MODE,
  LOCK_STATUS,
  LOCK_DATA
FROM performance_schema.data_locks;
```

2. Blocking vs Deadlock

Session 1 :

```
START TRANSACTION;
```

```
UPDATE employee
SET salary = 55000
WHERE id = 1;
```

Session 2 :

```
START TRANSACTION;
```

```
UPDATE employee
SET salary = 65000
WHERE id = 2;
```

Session 1 :

```
UPDATE employee
SET salary = 75000
WHERE id = 2;
```

Session 2 :

```
UPDATE employee
SET salary = 85000
WHERE id = 1;
```

```
SHOW ENGINE INNODB STATUS\G
```

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